The hook

Suppose I asked you to find the proteins driving severe COVID-19. Two methods compete. The first finds the proteins that best separate sick patients from healthy ones — the diagnostic markers. The second finds the proteins with the most connections in the cellular network — the hubs. Both are widely used. On the same patient cohort, they return entirely different answers. Not overlapping — *disjoint*.

But disagreement is not the deepest problem. **A protein elevated in disease may be elevated because of the disease, not because it drives the disease**. **Neither method distinguishes drivers from downstream signals. Six years since COVID ravaged the world, more than four hundred thousand papers, and no agreement on which is which.**

The question

Both methods miss the question that matters. **A protein with a strong individual signal may simply be reflecting damage — not a switch that caused it. A protein with many connections may be vital to cellular function but indifferent to *this* disease.** The right question is not which proteins are loud, or which proteins are central. **The right question is which proteins *drive* the disease. That question has a precise graph-theoretic form**. **Anchor the cascade at both ends — viral entry and immune activation at one end, clotting and organ damage at the other** — **and ask: what is the smallest set of intermediate proteins through which every signal must pass? That set is the minimum vertex cut.**

Why vertex cut, and not max-flow or edge cut? The three are computational siblings — they run on the same algorithm — but they answer different questions. Max-flow asks how much signal the cascade can carry. Edge cut asks which interactions are essential. Vertex cut asks which *proteins* are essential. Drugs bind to proteins, not to interactions. The medical question — and the answer therapeutic targeting can act on — lives at the vertex.

The mathematics behind this is a century old — Menger's theorem, 1927. **It guarantees that if you can exhibit a cut of a given size alongside the same number of disjoint paths through it, the cut is provably minimum. No solver, no statistical test. A certificate you can verify by hand.**

The result

So let me walk you through the poster.
[gesture to the pipeline figure] The **pipeline** shows the workflow: five public databases integrated into a single disease-specific graph of nearly a thousand proteins, vertex capacities derived from Welch t-statistics on 202 patients, and the cut recovered by max-flow.

[gesture to the methodology figure] **The methodology figure** shows the mathematical reduction. A toy graph in panel A. Node-splitting in panel B — each candidate protein splits into two halves joined by a finite-capacity arc. The scissors mark where max-flow saturates and the cut is found. And in the bottom row, the result made physical: on the left, the cascade is active, signal flowing from sources to sinks. On the right, the cut is removed — the same proteins now drawn as pale outlines — and the cascade is structurally silenced. The clinical pathology cannot form because the path no longer exists.

That is the method.   Now the answer.

*[**gesture to the hero visual and Table 1]*** **Applied to the COVID-19 interactome,** the cut has size six. The six proteins are listed in Table 1. Five are existing or investigational drug targets; two have FDA-approved drugs. The method recovers what the medical community arrived at independently — a sanity check that says the mathematics is finding the right kind of object. But the recovery of known targets is not the result. The result is what survives perturbation.
[gesture to Figure 5, bootstrap stability] We resampled the 202 patients a thousand times and recomputed the cut on each resample. One protein appears in *every* resample. A second appears in 990 of 1,000.

[gesture to Figure 6, Jaccard sensitivity] We then re-ran the analysis under three independent definitions of where the disease begins and ends. The cut is identical across all three.

[gesture to Figure 4, the three-way Venn]
And finally, the disjointness — the cut shares zero proteins with the centrality hubs, zero with **the statistical classifiers. Three a priori definitions of "disease-critical" coincide on no vertex, because they are measuring three different things, and only one of them is structural.**

The surprise

Five of the six cut proteins are already known. *[gesture to the highlighted row in Table 1]* The sixth — the highlighted row — has no prior COVID-19 literature. It is a lipid-binding protein with emerging neuroprotective interest in unrelated contexts, and the method places it in the bottleneck with 99% bootstrap stability.

What is important to think about is that the method, given no medical input, produced a structural prediction the literature has not. That is hypothesis-generating — the kind of result a wet lab can take to the bench.

And not in isolation. The cut is a *set* of six; the set is the prediction. Removing one cut vertex leaves five paths open. **The translational implication is combination intervention, not single-target drug discovery — which separates this work from most computational biomarker studies that deliver one protein at a time.**

The closer

Two final points.

First, this transfers. Any disease admitting an input → cascade → clinical-output decomposition is a candidate. Cancer signaling, autoimmune cascades, neurodegenerative pathways, sepsis. **Define the source set, define the sink set, compute the cut. The COVID-19 result is a demonstration; the methodology is the contribution.**

Second, the graph theory is a century old. Menger published in 1927. What is new is its application to high-dimensional clinical proteomics — a place where the graph theory has not been brought before, because the graph theorists have not worked on disease, and the disease researchers have not been familiar with the tool.

Centrality finds important nodes. Classification finds discriminative nodes. Minimum vertex cut finds *structurally indispensable* nodes. Three different questions, three different answers — and until now, the third has not been asked in clinical proteomics.

Thank you.